

ProductX Administration Guide

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Product: ProductX Enterprise Workflow Automation Platform
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Document Control Information

Field	Value
Document Type	Administration Guide
Review Cycle	Annual
Applicable Environments	Production, Staging, UAT, Development
Audience	Platform Administrators, Identity Administrators, Security Operations, Infrastructure Teams
Regulatory Relevance	ISO 27001, SOC 2 Type II, GDPR, HIPAA, PCI DSS

1. Purpose

This guide defines the administrative procedures, governance requirements, operational standards, and security controls required to manage ProductX environments across AeroNexus Technologies and authorized customer deployments.

The document serves as the authoritative source for:

- Tenant administration
- Identity and access management
- Authentication configuration
- Single Sign-On (SSO) integration
- System monitoring
- Security operations
- Audit management
- Backup administration

- Disaster recovery execution
 - Third-party integration governance
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2. Scope

This guide applies to:

- Enterprise Platform Operations
- Identity Governance Team
- Security Operations Center (SOC)
- Customer Tenant Administrators
- Managed Services Teams
- Infrastructure Operations

This guide does not cover:

- Product development procedures
 - Internal software engineering workflows
 - Source code deployment processes
 - Database schema modifications
 - Infrastructure provisioning standards
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3. ProductX System Architecture Overview

3.1 Platform Architecture

ProductX follows a multi-tenant cloud-native architecture deployed across multiple availability zones.

Core Platform Components

Identity Service

Responsible for:

- Authentication
 - Authorization
 - Session management
 - Identity federation
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Workflow Engine Cluster

Responsible for:

- Workflow execution
 - Business rule processing
 - Approval routing
 - Process orchestration
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Integration Hub

Responsible for:

- API communication
 - Connector management
 - Event processing
 - Data synchronization
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Notification Service

Responsible for:

- Email delivery
 - SMS notifications
 - Push notifications
 - Microsoft Teams integration
 - Slack integration
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Analytics Platform

Responsible for:

- Dashboard generation
 - Reporting
 - SLA calculations
 - Trend analysis
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Audit Service

Responsible for:

- Event collection

- Administrative activity logging
 - Security audit trails
 - Compliance reporting
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3.2 Logical Architecture

Client Layer

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Identity Layer

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API Gateway

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Workflow Services

↓

Integration Services

↓

Data Services

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Storage Layer

3.3 High Availability Design

Production environments are deployed using:

- Minimum 3 application nodes
- Multi-AZ database replication
- Active-active load balancing
- Geo-redundant backups

Target availability:

99.95% monthly uptime

4. Administrative Roles

4.1 Platform Administrator

Responsibilities:

- System configuration
- Tenant administration
- User lifecycle management
- Connector management

Maximum recommended allocation:

1 administrator per 500 active users

4.2 Identity Administrator

Responsibilities:

- SSO configuration
 - Federation management
 - MFA enforcement
 - User provisioning oversight
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4.3 Security Administrator

Responsibilities:

- Security policy management
 - Audit reviews
 - Threat monitoring
 - Compliance validation
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4.4 Reporting Administrator

Responsibilities:

- Analytics governance

- Report distribution
 - Data retention review
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5. User Provisioning

5.1 User Provisioning Methods

Supported methods:

Manual Provisioning

Used for:

- Emergency access
 - External contractors
 - Temporary users
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SCIM Provisioning

Supported providers:

- Microsoft Entra ID
- Okta
- Ping Identity
- OneLogin

Provisioning interval:

Every 15 minutes

API-Based Provisioning

Supported through:

- REST APIs
- Identity connectors

Maximum batch size:

10,000 users per synchronization cycle

5.2 User Lifecycle States

State	Description
Pending	Awaiting activation
Active	User can access platform
Suspended	Access temporarily disabled
Disabled	Login prohibited
Archived	Retained for compliance

5.3 Automated Deprovisioning

Users are automatically disabled when:

- Employment termination event received
- Contractor expiration date reached
- Identity provider deactivation detected

Deactivation SLA:

Within 30 minutes of source system notification

6. Role Management

6.1 Role-Based Access Control (RBAC)

ProductX uses hierarchical RBAC.

Access determination factors:

- Assigned role
 - Department
 - Business unit
 - Workspace
 - Data classification level
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6.2 Standard Roles

Role	Description
User	Standard business user
Workflow Designer	Workflow development
Business Administrator	Department administration
Auditor	Read-only audit access
Platform Administrator	Full tenant administration

6.3 Privileged Access Controls

Administrative accounts require:

- MFA
- Quarterly certification
- Session recording
- Just-In-Time elevation

Maximum elevation duration:

8 hours

6.4 Access Review Process

Review frequency:

Role Type	Review Frequency
Standard Users	Semi-annually
Administrators	Quarterly
Security Roles	Monthly

7. Authentication Configuration

7.1 Supported Authentication Methods

Local Authentication

Available only when explicitly enabled.

Requirements:

- Minimum 14-character password
 - Password history: 24 passwords
 - Password expiration: 90 days
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Federated Authentication

Supported protocols:

- SAML 2.0
- OpenID Connect (OIDC)
- OAuth 2.0

Recommended deployment model:

Federated Authentication Only

7.2 Multi-Factor Authentication

Supported factors:

- Authenticator applications
- Hardware security keys
- Push notifications
- SMS (not recommended)

Administrative accounts:

MFA mandatory

7.3 Session Controls

Setting	Default
Session Timeout	30 minutes
Absolute Session Lifetime	12 hours
Idle Timeout	15 minutes
Concurrent Sessions	5

8. Single Sign-On (SSO) Setup

8.1 SAML Configuration Procedure

Step 1

Obtain metadata XML from Identity Provider.

Step 2

Navigate:

Administration → Identity Management → Federation

Step 3

Upload metadata file.

Step 4

Configure:

- Entity ID
- Assertion Consumer Service URL
- NameID Format
- Signature Algorithm

Recommended algorithm:

SHA-256

Step 5

Enable test mode.

Step 6

Validate authentication with pilot users.

Step 7

Promote configuration to production.

8.2 Enterprise SSO Scenario

Scenario:

AeroNexus acquires a regional subsidiary with 1,200 employees.

Requirement:

Provide ProductX access without creating separate credentials.

Solution:

- Configure trust relationship with subsidiary Azure tenant.
- Enable SAML federation.
- Map employee attributes through SCIM.
- Apply role assignment rules.

Expected onboarding duration:

Less than 48 hours.

9. Integration Setup

9.1 Supported Integration Types

- REST API
- SOAP

- GraphQL
 - Webhooks
 - Message Queues
 - Database Connectors
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9.2 Enterprise Connectors

Certified connectors:

- Salesforce
 - SAP
 - ServiceNow
 - Jira
 - Workday
 - Oracle ERP
 - Microsoft Dynamics
 - Zendesk
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9.3 Integration Approval Process

Required before production deployment:

1. Security review
 2. Data classification review
 3. API risk assessment
 4. Business owner approval
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9.4 Integration Scenario

Scenario:

Purchase approvals require SAP synchronization.

Process:

1. Workflow approved.
2. ProductX invokes SAP connector.
3. Purchase Order created.
4. Confirmation returned.
5. Audit event logged.

Failure threshold:

3 consecutive API failures trigger connector suspension.

10. Monitoring and Operational Visibility

10.1 Monitoring Categories

Infrastructure Monitoring

Metrics:

- CPU utilization
 - Memory utilization
 - Storage consumption
 - Network latency
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Application Monitoring

Metrics:

- Workflow throughput
 - API response time
 - Connector health
 - Error rates
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Business Monitoring

Metrics:

- SLA compliance
 - Approval duration
 - Workflow completion rate
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10.2 Alert Thresholds

Metric	Warning	Critical
CPU	70%	85%
Memory	75%	90%
API Error Rate	2%	5%

Metric	Warning	Critical
Workflow Failures	10/hour	25/hour

10.3 Escalation Matrix

Severity	Response Time
Critical	15 minutes
High	1 hour
Medium	4 hours
Low	1 business day

11. Audit Logging

11.1 Logged Activities

ProductX records:

- Authentication events
- User creation
- Role modifications
- Workflow publication
- Configuration changes
- Connector changes
- Administrative actions

11.2 Audit Retention

Event Type	Retention
Authentication Logs	3 years
Admin Activities	7 years
Compliance Logs	10 years
Security Events	7 years

11.3 Tamper Protection

Audit logs are:

- Immutable
 - Cryptographically signed
 - Time synchronized via NTP
 - Replicated to secondary storage
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11.4 Audit Investigation Scenario

Incident:

Unauthorized workflow modification reported.

Investigation Steps:

1. Retrieve workflow history.
2. Review admin activity logs.
3. Correlate login events.
4. Verify change approvals.
5. Export evidence package.

Target investigation completion:

Within 24 hours.

12. Backup Policies

12.1 Backup Objectives

Recovery Point Objective (RPO):

15 minutes

Recovery Time Objective (RTO):

4 hours

12.2 Backup Types

Continuous Transaction Backups

Frequency:

Every 15 minutes

Full Backups

Frequency:

Daily

Retention:

35 days

Monthly Archive Backups

Retention:

7 years

12.3 Backup Encryption

Requirements:

- AES-256 encryption
 - Key rotation every 12 months
 - Separate key storage
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12.4 Backup Validation

Restore testing frequency:

Quarterly

Success threshold:

100% restoration validation

13. Disaster Recovery Procedures

13.1 Recovery Scenarios

Availability Zone Failure

Expected recovery:

Less than 30 minutes

Regional Cloud Outage

Expected recovery:

Less than 4 hours

Data Corruption Event

Expected recovery:

Less than 2 hours

13.2 Disaster Recovery Workflow

Step 1:

Declare incident.

Step 2:

Activate DR team.

Step 3:

Failover services.

Step 4:

Validate platform health.

Step 5:

Restore integrations.

Step 6:

Business validation.

Step 7:

Return to normal operations.

13.3 Annual DR Testing

Requirements:

- Full failover exercise
 - Data restoration validation
 - Executive review
 - Lessons learned report
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14. Security Recommendations

Identity Security

- Enforce MFA universally.
 - Disable inactive accounts after 45 days.
 - Eliminate shared administrator accounts.
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Access Governance

- Conduct quarterly role reviews.
 - Remove excessive permissions.
 - Use approval-based access requests.
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Integration Security

- Rotate API credentials every 90 days.
- Restrict IP access where supported.
- Enable mutual TLS for critical integrations.

Audit Security

- Forward logs to SIEM.
 - Monitor privileged activities.
 - Review failed login patterns weekly.
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15. Compliance Requirements

ProductX administrative operations must comply with:

- ISO 27001
- SOC 2 Type II
- GDPR
- HIPAA
- PCI DSS
- NIST 800-53

Administrative changes impacting production require:

- Change record approval
 - Risk assessment
 - Rollback plan
 - Post-implementation review
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16. Exceptions Management

Exceptions may be granted for:

- Regulatory obligations
- Customer contractual commitments
- Emergency business continuity requirements

Approval authority:

Exception Type	Approver
Security Control	CISO Delegate
Authentication Policy	Identity Governance Director
Backup Policy	Infrastructure Director
Enterprise Standard	Executive Technology Council

Maximum exception duration:

180 days

Renewal requires reapproval.

17. Best Practices

Administration

- Maintain separate admin accounts.
- Use delegated administration where possible.
- Document all production changes.

Security

- Enable conditional access policies.
- Restrict privileged access by location.
- Review privileged sessions weekly.

Monitoring

- Establish baseline metrics.
- Investigate anomalies immediately.
- Automate alert routing.

Disaster Recovery

- Test failover regularly.
 - Validate backup integrity.
 - Maintain current runbooks.
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18. Frequently Asked Questions

Q1. How many administrators should manage a tenant?

Recommended ratio is one administrator per 500 active users.

Q2. Is local authentication recommended?

No. Federated authentication with MFA is the preferred configuration.

Q3. How quickly are users provisioned through SCIM?

Typically within 15 minutes.

Q4. Can ProductX support multiple identity providers?

Yes. Multiple SAML and OIDC providers may be configured simultaneously.

Q5. What is the maximum role inheritance depth?

Five levels of inheritance.

Q6. How long are administrative audit logs retained?

Seven years by default.

Q7. What happens if an integration continuously fails?

After three consecutive failures, connector health enters warning status. After ten failures within one hour, automatic suspension occurs.

Q8. How often should backup restoration testing occur?

Quarterly at minimum.

Q9. Are audit logs editable?

No. Audit records are immutable.

Q10. What is the supported disaster recovery objective?

RPO: 15 minutes

RTO: 4 hours

19. Reference Documents

Document ID	Document Name
ANX-PX-SEC-014	ProductX Security Administration Standard
ANX-PX-OPS-009	ProductX Operations Handbook
ANX-PX-UG-001	ProductX User Guide
ANX-IDM-005	Enterprise Identity Governance Policy
ANX-DRM-011	Data Retention Management Standard
ANX-BCP-017	Business Continuity and Disaster Recovery Framework
ANX-CHG-003	Enterprise Change Management Procedure

20. Approval Record

Role	Department	Status
Director, Enterprise Platform Operations	Enterprise Platforms	Approved
Director, Information Security	Security & Risk	Approved
Head of Identity Governance	Corporate Security	Approved
VP, Enterprise Platforms	Technology Operations	Approved

Document End